The first dissenter and the first contact: what the onset of disruption reveals about the ejected body

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We examine whether the onset of the disruption of a three-body system reveals which body will be ejected. Starting from the perturbed Lagrange triangle, we define the first dissenter in four ways and find, in 4,400 regularized planar integrations, that it does not predict the ejected body: with equal masses, the success rate lies between 29% and 40%, at the level of chance, because 105 to 135 close encounters occur between the onset and the ejection; with unequal masses, the apparent signal is an effect of the mass. We then test, in 30,000 integrations with masses 1, 1, and q (q from 2 to 30) and three types of start, whether the light mass that first forms a close pair with the heavy one remains with it. In the symmetric start, this occurs in 64% to 83% of cases, with an advantage that grows with q, whereas initial proximity predicts nothing. In random starts, the effect falls to 54% to 61% and is largely confounded with the advantage of starting closer. An energy decomposition, in 18,000 additional integrations, shows the origin of the effect: when the first contact pair already holds more binding than the total energy of the system, the other light mass is expelled in the encounter itself and the light mass of the contact remains in 91% to 100% of cases; outside these cases, the effect is smaller and grows with the binding of the pair.

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I. INTRODUCTION

The equilateral Lagrange triangle with three equal masses is an exact solution of the three-body problem, but it is unstable: it does not satisfy the Gascheau and Routh condition, which requires a dominant body [1, 7]. Any perturbation, however small, grows until it breaks up the triangle, and the system evolves toward the ejection of one of the bodies. Note FR-TC-05 showed that, with equal masses, the surviving binary is only determined at the last close encounter [6]. This note examines the same question from the other end, the onset of the disruption, with two questions: is the body that first departs from the configuration, the "first dissenter", the one that will be ejected? And, when two light masses compete for a heavy one, does the light mass that first forms a close pair with the heavy one tend to remain with it?

II. METHOD

A. Simulations

We use the regularized integrator of the previous notes [5], in planar simulations with $G = 1$, without softening of the potential and with the ejection criterion of note FR-TC-05 [6]. The relative energy error remained below 5×10^{-9} in all runs. The integration limit was 500 crossing times.

B. First dissenter

We start from the equilateral triangle of side 1 rotating rigidly with angular velocity $\Omega = \sqrt{GM}$, with independent Gaussian perturbations of amplitude ε in position and velocity. For equal masses, we use $\varepsilon = 10^{-8}$, 10^{-5}, and 10^{-3}, with 400 runs each. For unequal masses, we use the groups (1, 1, 3), (1, 1, 10), (3, 3, 1), and (10, 10, 1), with $\varepsilon = 10^{-5}$ and 10^{-3} and 400 runs each. The dissenter was defined in four ways: the body that received the largest position perturbation; the body farthest, relative to equilibrium, from the center of mass of the other two, when the maximum deformation of the sides reaches 10% and 30%; and the body excluded from the first close encounter, that is, the first instant at which a side falls below 0.2.

C. First contact

For masses $(1, 1, q)$, with $q = 2$, 3, 10, and 30, we call first contact the first instant at which the distance between the heavy mass and one of the light ones falls below a fraction f of the characteristic radius $r_v = \sum_{i<j} m_i m_j / |W_0|$, where W_0 is the initial potential energy; we use $f = 0.1$, 0.2, and 0.3. We tested three types of start, with 2,500 runs in each combination, for a total of 30,000: the Lagrange triangle with $\varepsilon = 10^{-3}$, in which the two light masses start in equivalent positions; the start from rest with random positions; and the start with random velocities and virial ratio 0.5. As a control, we evaluated the predictor "the light mass that started closest to the heavy one remains", which separates the effect of contact from the effect of the initial geometry. We consider only the runs in which one of the light masses was ejected.

D. Pair binding at first contact

To investigate the origin of the effect, we repeated 1,500 runs per combination of start and mass ratio, with the same

seeds, for a total of 18,000, and recorded, at the instant of first contact with $f = 0.1$, the binding energy of the newly formed pair, $E_b = -E_{ij}$, in units of the magnitude of the total energy of the system, $|E|$. The case $E_b > |E|$ has a direct meaning: the pair holds more binding than the whole system, so that, to first approximation, the energy left for the third body is positive.

III. RESULTS

A. Equal masses: the dissenter does not predict the outcome

Table I shows that none of the four definitions of the dissenter predicts the ejected body better than chance. Between the first deformation and the ejection there are, in the median, from 105 to 135 close encounters. The initial phase, in contrast, follows the linear analysis: the time to 10% deformation, measured in revolutions of the triangle, was 1.05, 2.09, and 3.63 for $\varepsilon = 10^{-3}$, 10^{-5}, and 10^{-8}. Each thousandfold reduction in the perturbation delays the breakup by slightly more than one revolution, consistent with a growth rate of order Ω.

B. Unequal masses: the dissenter signal is the mass in disguise

With unequal masses, the lightest body was ejected in 94% to 100% of the runs. The "farthest" dissenter was correct in 32% to 52% of cases, above chance in the groups with two large masses, but only because, in these groups, the dissenter is often the light body. In the groups with two equal light masses, the dissenter does not indicate which of the two is ejected: the success rate was between 43% and 52%, against 50% for chance. The median number of close encounters before ejection fell from 65 to 93 in the groups with two light masses to 22 to 33 in the (3, 3, 1) group and 4 in the (10, 10, 1) group.

C. First contact

Table II summarizes the effect of first contact with $f = 0.1$, and Fig. 1 shows it for the symmetric start at the three thresholds.

In the symmetric start, the effect is strong and grows with the mass ratio, from 64% for $q = 2$ to 83% for $q = 30$, whereas initial proximity has no predictive power (48% to 51%). The effect is stronger for tighter contacts: with $f = 0.3$, it lies between 53% and 64%. In the random starts, the light mass of the first contact remains in 54% to 61% of the runs, but the initial proximity predictor is correct almost as often, because the light mass that starts closest is generally the one that makes the first contact. In the discordant cases, contact wins in the start from rest for $q \leq 10$ (60% to 71%) and ties in the virial start (49% to 52%). When the first contact occurs after one crossing time, in the virial start, the effect rises to 54% to 74%, with smaller samples.

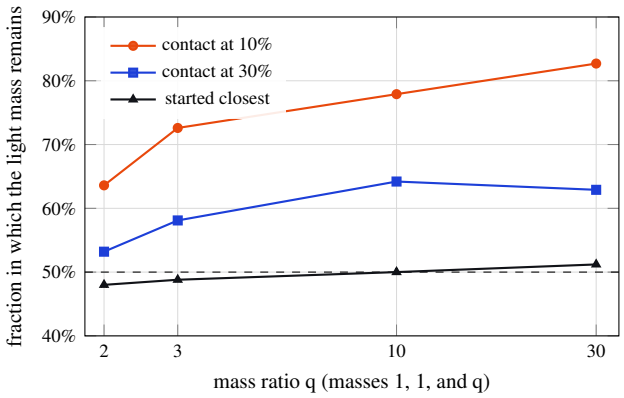


Fig. 1. Symmetric start (perturbed Lagrange triangle): fraction of runs in which the light mass of the first contact remains with the heavy one, for contact at 10% and at 30% of the characteristic radius, and fraction in which the light mass that started closest remains, as a function of the mass ratio q. The dashed line indicates chance.

D. Origin of the effect

Table III separates the runs in which the first contact pair is born with $E_b > |E|$. In these cases, the light mass of the contact remains in 91% to 100% of the runs, in the cells with at least 50 cases, and the ejection of the other light mass occurs, in 88% to 100% of them, less than one crossing time after the contact: the first contact is itself the encounter that expels. These cases are frequent in the symmetric start (20% to 36% of the runs), less frequent in the virial start (11% to 16%), and rare in the start from rest (up to 4%), which explains much of the difference between the families.

With the decisive contacts excluded, a smaller effect remains. In the symmetric start, the light mass of the contact remains in 56% to 74% of the runs, still growing with q; in the start from rest, in 56% to 59%; in the virial start, in 47% to 52%, at the level of chance. Within this residual, binding still matters: in the most bound third of pairs, retention rises to 63% to 80% in the symmetric start and to 55% to 66% in the virial start, but not in the start from rest. We also recorded the ratio between the orbital velocity of the pair and the arrival velocity of the other light mass, as in note FR-TC-06. At the instant of contact, this ratio is a misleading measure: it mainly reflects the velocity of the other light mass, which is high precisely when it is on its way out.

IV. DISCUSSION

With equal masses, the absence of memory between the onset and the end of the disruption is a consequence of the long sequence of close encounters that separates them: each encounter redistributes energy and angular momentum among the pairs and erases the information about the initial deformation. The result complements that of note FR-TC-05 [6]: neither the beginning nor the middle of the evolution predicts the outcome with equal masses.

Table I. Equal-mass Lagrange triangle: success rate for the ejected body for each definition of the dissenter. Chance: 0.33. Ejections per column: 346, 349, and 348.

Definition of the dissenter	$\varepsilon = 10^{-8}$	$\varepsilon = 10^{-5}$	$\varepsilon = 10^{-3}$
largest initial perturbation	0.33	0.33	0.38
farthest at 10% deformation	0.32	0.33	0.31
farthest at 30% deformation	0.29	0.35	0.34
excluded from the first close encounter	0.40	0.33	0.38

Table II. Fraction of runs in which the light mass of the first contact with the heavy one remains in the final binary ($f = 0.1$), compared with the initial proximity predictor. "Discordant": runs in which the two rules point to different light masses, and fraction in which contact wins. Chance: 0.50. The 95% confidence interval is about ± 0.02 in the main columns.

Start	q	Runs	Contact remains	Closest remains	Discordant	Contact wins
Lagrange	2	1819	0.64	0.48	980	0.64
Lagrange	3	1743	0.73	0.49	949	0.72
Lagrange	10	1206	0.78	0.50	651	0.77
Lagrange	30	659	0.83	0.51	319	0.82
rest	2	2346	0.61	0.59	152	0.66
rest	3	2392	0.58	0.55	140	0.71
rest	10	2440	0.55	0.54	117	0.60
rest	30	2388	0.57	0.57	68	0.47
virial	2	1886	0.55	0.53	791	0.52
virial	3	1928	0.54	0.55	819	0.49
virial	10	1832	0.59	0.59	810	0.51
virial	30	1796	0.59	0.60	754	0.50

The first contact effect has a direct physical interpretation. When a light mass comes very close to the heavy one, a pair forms whose binding energy grows with the mass of the heavy body. By Heggie's law [2, 3], later encounters with the other light mass tend to harden this hard pair and to fling out the intruder, so that the advantage of arriving first increases with q and with the closeness of the contact. In the symmetric start, the advantage appears in isolation, because the two light masses start in equivalent positions. In the random starts, it mixes with the initial geometry and, in the virial start, the initial velocities distribute the approaches in such a way that the first contact loses information, except when it occurs late.

The decomposition in Table III shows that much of the first contact effect comes from decisive contacts, in which the pair formed is already hard enough for the third body to have escape energy: the expulsion happens in the encounter itself. The residual effect is consistent with Heggie's law [2, 3]: more tightly bound pairs tend to survive the subsequent encounters and to eject the intruder. The relevant measure is the binding energy of the pair compared with the total energy of the system, and not the instantaneous velocity ratio.

The main limitations are the following. The simulations are planar. The definitions of dissenter, contact, and close encounter are our own choices, although the results are stable across the thresholds tested. The symmetric start is a special case: the strong effect it reveals does not carry over fully to generic initial conditions. Contact detection uses the recorded trajectory, which is decimated in long runs, and this may miss very brief approaches.

V. CONCLUSIONS

The first body to depart from the Lagrange triangle does not predict the ejected body: with equal masses, the success rate is at the level of chance, and, with unequal masses, the apparent signal is an effect of the mass. The first tight contact between a light mass and the heavy one, on the other hand, has a real effect: in a symmetric start, the light mass that forms it remains with the heavy one in 64% to 83% of cases, with an advantage that grows with the mass ratio. In generic starts, the effect is weak and largely equivalent to the advantage of starting closer. The decomposition by pair binding shows the origin of the effect: in 11% to 36% of the runs with a velocity or symmetric start, the first contact pair already holds more binding than the total energy of the system, and the other light mass is expelled in the encounter itself, with the light mass of the contact remaining in 91% to 100% of cases; outside these cases, the effect is smaller and grows with the binding of the pair.

Table III. First contact decomposed by the binding of the newly formed pair. "Decisive": runs with $E_{\rm b} > |E|$ at contact. "Others": the remaining runs, with the fraction in which the light mass of the contact remains, overall and in the most bound third of pairs. Chance: 0.50.

Start	q	Runs	Decisive	Contact remains (decisive)	Contact remains (others)	Others, most bound third
Lagrange	2	1108	0.20	0.99	0.56	0.63
Lagrange	3	1046	0.26	0.99	0.62	0.71
Lagrange	10	724	0.36	1.00	0.66	0.72
Lagrange	30	396	0.28	1.00	0.74	0.80
rest	2	1396	0.04	0.91	0.59	0.57
rest	3	1439	0.04	0.96	0.57	0.58
rest	10	1468	0.01	0.60	0.56	0.60
rest	30	1428	0.00	no cases	0.56	0.57
virial	2	1127	0.11	0.91	0.49	0.55
virial	3	1167	0.14	0.92	0.47	0.60
virial	10	1096	0.15	0.96	0.52	0.66
virial	30	1072	0.16	0.97	0.52	0.60

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

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